

Amal Jyothi College of Engineering (Autonomous)

Amal Jyothi Scholars Community for Inspiring Innovation (ASCII)

R & D arm of IEDC

19th ASCII- IÅ²U-2025 Contest test

1.TITLE OF THE PROJECT: TITLE OF THE PROJECT: Careerist - AI-Powered Career Companion for Students & Graduates

2. PROJECT SYNOPSIS (100 WORDS):

Careerist is an AI-powered platform that serves as a human-like career companion for confused students and graduates navigating their futures. At its core is our advanced AI chatbot that deeply understands each user's personality, strengths, and aspirations, providing genuinely personalized life guidance. The platform features dynamic dashboards that evolve with user progress: Student Dashboard for academic and career discovery, Graduate Dashboard for professional advancement. Our machine learning algorithms enhance interview skills, communication abilities, and workplace readiness through real-time analysis and feedback. By combining empathetic AI understanding with comprehensive career tools, Careerist transforms overwhelming career confusion into confident, well-informed decisions for lifelong success.

3. POTENTIAL APPLICATIONS:

Educational Institutions:

- **High schools for career counseling and college preparation guidance**
- **Universities for student career services and graduate placement programs**
- **Community colleges for academic planning and transfer preparation**
- **Online educational platforms for enhanced student career support**
- **Academic advisors for personalized student guidance**
- **Career centers for comprehensive student services**

Individual Students:

- **High school students exploring career paths and college options**
- **College students planning academic majors and career directions**
- **University seniors preparing for post-graduation transitions**
- **Community college students navigating transfer or workforce entry**
- **Students uncertain about their academic and career choices**

Recent Graduates:

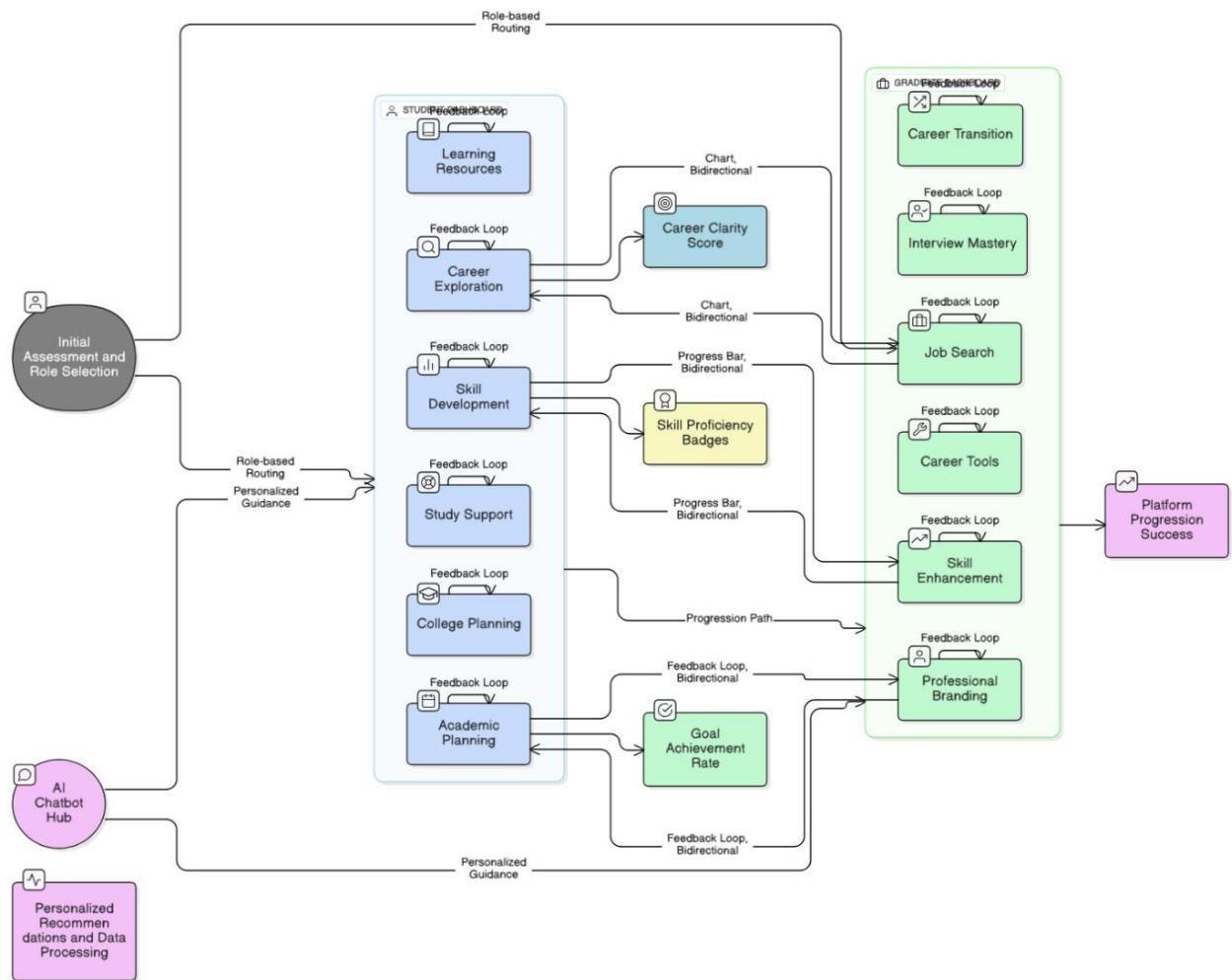
- **New graduates seeking first professional positions**
- **Recent alumni transitioning from education to employment**
- **Young professionals in early career development**
- **Graduates changing career directions or industries**
- **International students navigating local job markets**

Personal Use Cases:

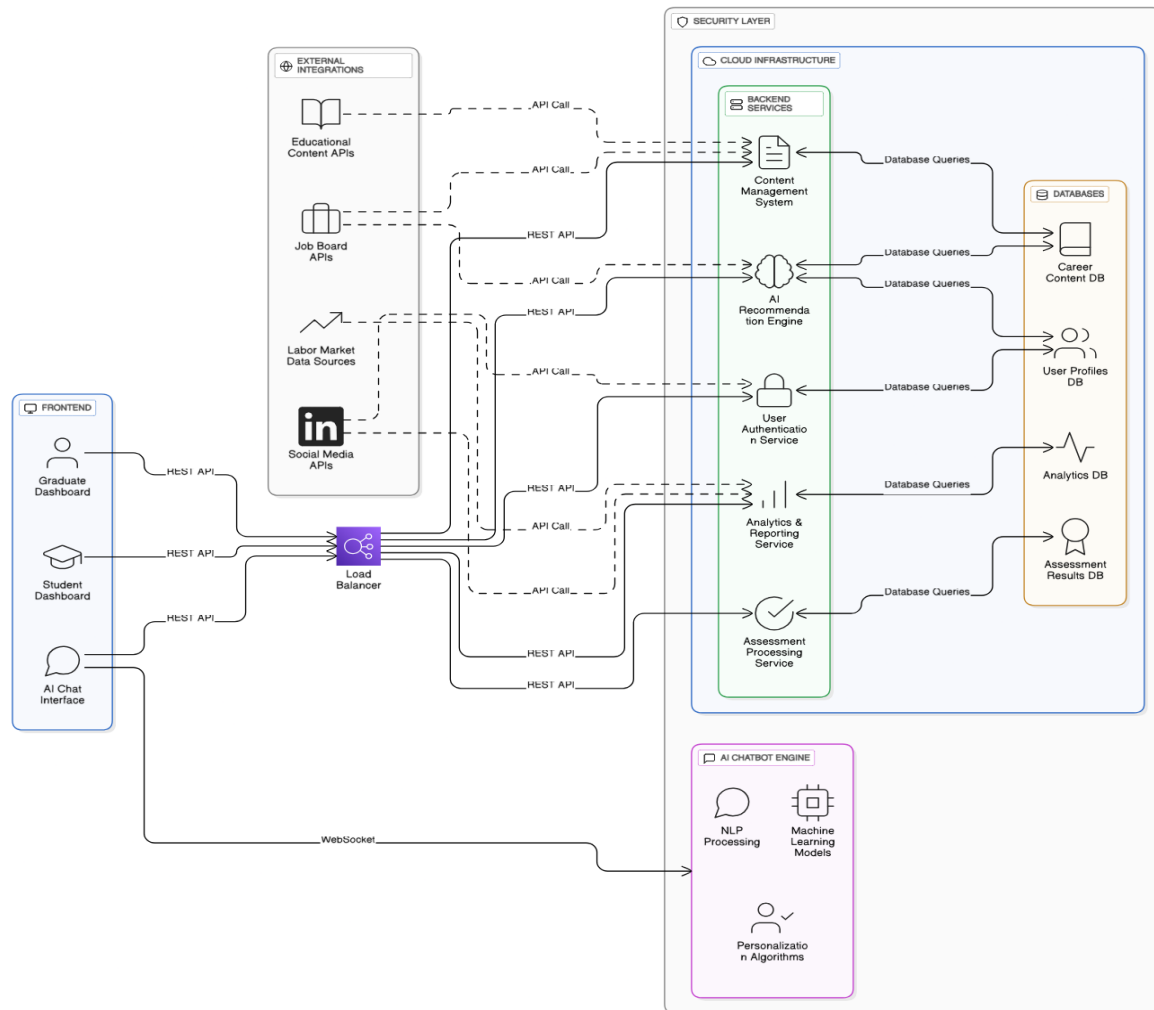
- **Individual career exploration and self-discovery**
- **Personal skill development and professional growth**
- **Academic and career goal setting and tracking**
- **Interview preparation and professional development**
- **Long-term career planning and decision-making**

4. GENERAL PLAN OF IMPLEMENTATION (Block diagram, flow chart etc.):

FLOW CHART:



BLOCK DIAGRAM:-



5. PROJECT TIMELINE WITH MILESTONES:

Phase 1: Foundation & Data Setup (Months 1-2)

- Complete system architecture and database design
- Set up data collection infrastructure for ML training
- Develop user authentication and role-based routing system
- Create initial assessment questionnaire and role selection logic
- **Milestone:** Core platform infrastructure and data pipeline operational

Phase 2: AI Core & ML Model Development (Months 3-4)

- Implement natural language processing engine
- **Develop and train ML models for personality assessment and user profiling**
- **Create interview analysis ML algorithms (speech, communication patterns)**
- **Train recommendation models using career and educational datasets**
- Build basic chatbot interface and conversation flow
- **Milestone:** Core AI/ML models trained and functional

Phase 3: Student Dashboard & ML Integration (Months 5-6)

- Develop career exploration and subject discovery modules
- **Implement ML-powered academic performance prediction**
- **Train models for personalized study recommendations**
- Create college and major exploration with ML-based matching
- **Milestone:** Student features with integrated ML recommendations

Phase 4: Graduate Dashboard & Advanced ML (Months 7-8)

- Build job search and application tracking systems
- **Develop ML-powered interview simulation and feedback system**
- **Train communication analysis models for real-time coaching**
- **Implement ML resume optimization and job matching algorithms**
- Create networking and professional development features
- **Milestone:** Graduate features with advanced ML interview coaching

Phase 5: ML Model Refinement & Cross-Platform Intelligence (Months 9-10)

- **Continuous ML model training with user interaction data**
- **Implement deep learning for human-like conversational AI**
- **Develop predictive analytics for career success patterns**
- Create dynamic content personalization using ML insights
- **Milestone:** Fully intelligent, self-improving AI system

Phase 6: ML Optimization & Launch (Months 11-12)

- **Fine-tune ML models based on beta user data**
- **Implement real-time model updates and A/B testing**
- Conduct comprehensive testing with target institutions
- Launch platform with continuously learning AI system
- **Milestone:** Production-ready platform with adaptive ML capabilities

6. EXPECTED OUTCOMES OF THE PROJECT:

For Students:

- 85% improvement in career clarity and direction among confused students
- 70% increase in informed college and major selection decisions
- Enhanced academic performance through AI-powered study recommendations
- Significant improvement in interview skills and professional communication
- Comprehensive skill development portfolio with measurable progress tracking

For Recent Graduates:

- 60% faster job placement rates compared to traditional methods
- Improved interview success rates through ML-powered coaching and feedback
- Professional branding optimization leading to better job opportunities
- Reduced career transition anxiety through personalized guidance
- Higher job satisfaction through better career-role matching

For Educational Institutions:

- Enhanced career counseling capabilities with data-driven insights
- Improved student retention and graduation rates

- Better alumni employment outcomes and institutional reputation
- Streamlined career services operations with AI automation
- Comprehensive analytics on student career development patterns

Platform Capabilities:

- Fully functional AI chatbot with human-like understanding and empathy
- Dynamic, personalized user experiences that evolve with progress
- Real-time ML analysis for interview improvement and skill development
- Comprehensive database of career pathways and market trends
- Scalable platform serving thousands of students and graduates simultaneously

Measurable Impact:

- 90% user satisfaction rate with AI-powered guidance
- Demonstrable skill improvement through before/after assessments
- Significant reduction in career-related stress and uncertainty
- Increased user engagement with career planning and development activities

7. BUDGETARY REQUIREMENTS* (components & approximate costs in tabular form) (*No funds for travel and equipment is allotted)

Category	Component	Monthly Cost (₹)	3-Month Cost (₹)	Purpose
Cloud Infrastructure	AWS Free Tier + Basic	₹800	₹2,400	Platform hosting and testing
	Database Services	₹500	₹1,500	User data storage
	CDN & Storage	₹300	₹900	Content delivery
AI/ML Services	OpenAI API (Dev Credits)	₹1,200	₹3,600	Chatbot development
	Speech Analysis APIs	₹400	₹1,200	Interview features testing
Development Tools	GitHub Pro	₹200	₹600	Code repository
	Development Environment	₹300	₹900	Development tools
Third-party APIs	Job Board APIs (Basic)	₹500	₹1,500	Job market data
	Educational Content API	₹300	₹900	Learning resources
Security & Monitoring	SSL & Basic Monitoring	₹150	₹450	Security and tracking
Miscellaneous	Domain & Email	₹100	₹300	Professional setup
	Testing Services	₹200	₹600	Quality assurance
	Documentation Tools	₹150	₹450	Project documentation
TOTAL		₹5,100	₹15,300	Complete development setup